

## Article

# Downscaling of Urban Land Surface Temperatures Using Geospatial Machine Learning with Landsat 8/9 and Sentinel-2 Imagery

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## Abstract

Urban surface temperatures are increasing because of climate change and rapid urbanisation, contributing to the urban heat island (UHI) effect and significantly influencing local climates. Satellite-derived land surface temperature (LST) plays a vital role in analysing urban thermal patterns. However, current satellite thermal infrared (TIR) sensors have a low spatial resolution, making it difficult to accurately capture the complex thermal variations within urban areas. This limitation affects the assessments of UHI effects and hinders effective mitigation strategies. We proposed a hybrid model named “geospatial machine learning” (GeoML) to address these challenges, combining random forest and kriging downscaling techniques. This method utilises high spatial resolution data from Sentinel-2 to enhance the LST derived from Landsat 8/9 data. Tested in Perth, Australia, GeoML generated an enhanced LST with good agreement with ground-based measurements, with a Pearson’s correlation coefficient of 0.85, a root mean square error (RMSE) of 2.7 °C, and a mean absolute error (MAE) of less than 2.2 °C. Validation with LST derived from another TIR sensor also provided promising outputs. The results were compared with the high-resolution urban thermal sharpener (HUTS) downscaling methods, which GeoML outperformed, demonstrating its effectiveness as a valuable tool for urban thermal studies involving high-resolution LST data.

**Keywords:** land surface temperature; downscaling; urban areas; random forests; kriging; ground-based data; validation



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## 1. Introduction and Previous Work

Land surface temperature (LST), a measure of the temperature of the Earth’s surface at a specific location, is an important variable in understanding climate change, precision agriculture, forest fire detection, soil moisture, and environmental monitoring at both local and global scales. Traditional temperature measurements relied on meteorological stations installed at sparse points worldwide. Due to the advent of satellite technology and data processing, temperature trends and weather forecasts can be made using constellations of

satellites around the globe. Thermal infrared (TIR) remote sensing offers extensive coverage and repeatability, making it a cost-effective tool for detecting temperature variations across diverse landscapes [1]. Satellite Earth observation of LST has become an important source of information for studying heatwaves and associated overheating of the urban landscape. The study of the urban heat island (UHI) effect has become highly important nowadays, as it affects human health, well-being, and energy consumption, given that more than half of the world's population currently lives in cities. However, the coarse spatial resolution of TIR sensors limits their applicability in urban settings, where fine-scale land cover variations significantly influence LST patterns [2].

TIR remote sensing has been used to map the LST over extended areas for several decades. TIR radiation has longer wavelengths than visible and near-infrared (VNIR), where the emitted radiation has lower energy than reflected optical bands. This necessitates larger detector elements to capture sufficient energy to meet the signal-to-noise ratio requirements, resulting in coarser spatial resolution than the VNIR, as the sensors must view large areas of the Earth's surface to obtain detectable TIR radiation [3]. While coarse-resolution LST data can effectively capture urban–rural temperature differences, it fails to resolve fine-scale temperature variations within heterogeneous urban environments, for instance, when studying the heat behaviour of roof materials in urban settlements. This limitation leads to thermal mixture effects, where multiple land cover types contribute to the temperature of a single pixel, reducing the accuracy of LST retrievals [4]. Consequently, enhancing the spatial resolution of LST is essential for accurately assessing the UHI effects, urban energy efficiency, and heat vulnerability at local scales. According to a survey conducted by [5], high spatial resolution LST with a maximum threshold of 15 m is necessary to study energy efficiency and urban planning. Higher spatial resolution thermal data are needed to enable the discrimination of fine-scale features like green spaces, water bodies, and buildings and their corresponding temperatures. Moreover, extreme heat events have risen in the last few decades, making urban centres significantly hotter than surrounding rural areas. To accurately identify vulnerable areas for UHI, a measure of LST at a fine scale is necessary [6].

To overcome these challenges, various LST downscaling techniques have been developed, generally categorised into physical and statistical approaches [7]. Physical downscaling approaches are based on subpixel thermal anomaly and emissivity using the concept of surface energy balance and physical modelling of solar radiation (e.g., [8–10]), but these methods are complex and computationally intensive and require extensive ancillary data, making them less commonly used. In contrast, statistical downscaling exploits empirical relationships between coarse-resolution LST and high spatial resolution auxiliary variables, such as spectral indices and surface albedo, to estimate the high spatial resolution LST (e.g., [11,12]). Statistical downscaling methods offer a more efficient and widely adopted approach despite some limitations, such as the lack of a physical mechanism in the downscaling method and the issue of extension of LST downscaling to other study areas due to empirical regressive relationships [13]. Early statistical downscaling methods, such as TsHARP [14] and DisTrad [15], relied on simple linear and quadratic relationships with vegetation indices, mainly the normalised difference vegetation index (NDVI). Still, their accuracy is limited in urban areas due to the complex, non-linear interactions between LST and land cover characteristics. Several subsequent studies (e.g., [16,17]) demonstrate that NDVI cannot explain the variation of LST, especially in urban areas where vegetation is sparse and filled with different materials such as roofs, paving, and more. As such, more sophisticated models are needed to manage the complexities associated with LST variability effectively.

To address these challenges, Dominguez et al. [18] proposed the high-resolution urban thermal sharpener (HUTS) to downscale LST derived from Advanced Spaceborne Thermal Emission and Reflection Radiometer (ASTER) data for urban applications. This model is based on a fourth-order non-linear regression between the NDVI and the surface albedo. This approach generated more accurate high-resolution LST compared to TsHARP and DisTrad [18]. However, the accuracy of the method has not been examined within different land cover classes in urban areas. Recent advancements in machine learning (ML) have introduced more robust non-linear downscaling techniques. Models such as random forests (RF), support vector machines (SVM), and artificial neural networks (ANN) have demonstrated superior performance in capturing complex LST patterns [19,20]. RF, in particular, has gained popularity due to its ability to handle high-dimensional data, resilience to outliers, and robust variable selection [21]. Studies have shown that RF-based downscaling achieves higher accuracy than traditional statistical methods (e.g., [22,23]). Wang et al. [24] utilised RF in conjunction with multiple remote sensing indices to downscale LST in arid regions, achieving significant improvements in accuracy. However, ML methods alone do not explicitly account for spatial autocorrelation, which is a crucial factor in LST distribution [3].

To overcome this limitation, geostatistical downscaling methods such as kriging have been applied to interpolate LST at finer scales while preserving the LST spatial dependency [25]. Kriging variants, including ordinary kriging and cokriging, use spatial interpolation to estimate LST at high resolutions by incorporating observed temperature patterns and high spatial resolution temperature predictors [25–27]. Consequently, kriging techniques consider the spatial variability of LST across different land cover types, resulting in more accurate downscaling results. In addition, apart from the downscaled LST, kriging also provides an estimate of uncertainty at each pixel, allowing a better understanding of the confidence in the downscaled data [28]. Nevertheless, kriging alone struggles with high-dimensional feature spaces and non-linear LST–land cover relationships [26]. A promising solution lies in the integration of machine learning and geostatistical downscaling methods, leveraging the strengths of both approaches. Hybrid methods combine ML’s predictive power with kriging’s spatial interpolation to enhance downscaling accuracy. Specifically, ML can estimate the trend component of LST while kriging refines the residuals, capturing spatial dependencies that ML methods typically overlook [22]. This hybrid approach of ML and geostatistical methods presents a novel framework for LST downscaling that has not been extensively explored in the literature.

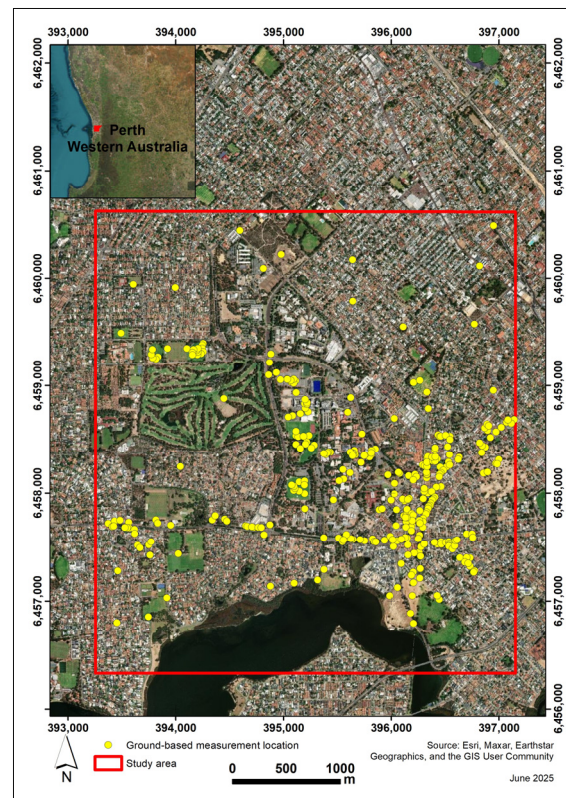
In this paper, we aim to fill the gap by combining two existing approaches: the random forest (RF) method to determine the high-resolution LST trend and the kriging technique to downscale the LST residual to obtain more accurate final enhanced LST results. This method is named “geospatial machine learning” or GeoML for the rest of the paper, as it is a hybrid modelling using machine learning and ordinary kriging to downscale LST for urban applications. By combining both methods, we aim to achieve more accurate LST retrievals in complex urban environments. The rest of this paper is organised as follows. Section 2 provides the materials and methods, introducing our study area and datasets. Section 3 describes the results and validations of the downscaled LST, Section 4 discusses the main findings, and Section 5 concludes the paper.

## 2. Materials and Methods

### 2.1. Study Area

The presented downscaling approach is tested on the Curtin University Bentley campus and in the surrounding areas, located in the City of Canning, Western Australia. It is approximately 16 square kilometres (3.8 km × 4.2 km) (Figure 1). The local climate is

classified as warm temperate and Mediterranean, where it is hot, dry, and sunny during the summer and milder and wetter during the winter [29]. Recently, the area of interest has experienced a phase of urban infill [30]. It can be considered a medium-density building in urban areas. The area has plenty of land cover types, including buildings where the roof type is mainly tiles and Colorbond (steel), vegetation covers such as trees and grasslands, paved areas such as parking lots, wide and narrow streets, open earth areas, and water bodies to demonstrate the variability of the UHI effect. Outside the campus, the zone encompasses residential block areas bounded by roads. A part of the Canning River flows in the southern part of the study area (Figure 1).



**Figure 1.** Location map of the study area and positions of the ground-based temperature measurements.

## 2.2. Data

### 2.2.1. Satellite Data

Spatial downscaling was performed using multispectral satellite imagery data collected by the Landsat and Sentinel missions. Landsat 8 (L8) and 9 (L9) have two TIR bands with 100 m spatial resolution covering wavelengths between 10.60 and 12.5  $\mu\text{m}$ . L8 and L9 acquire images of the Earth's surface every 16 days, with an 8-day offset between their acquisition cycles. The study area benefits from an intersection of L8 and L9 orbits, providing two satellite images every 8 days. The data for L8/L9 are directly accessible within the Google Earth Engine (GEE) repository as atmospherically corrected surface reflectance and surface temperature Level 2, Tier 1 products, with an overpass time between 10:05 and 10:11 a.m. local time.

Sentinel-2 (S2) features 13 bands ranging from 0.40 to 2.28  $\mu\text{m}$ , of which 11 bands have 10 m spatial resolutions. The temporal resolution of the S2 satellites is 5 days. S2 does not have TIR bands. Only six bands corresponding to the visible and near infrared (VNIR) and shortwave infrared (SWIR) bands were employed for this study. The atmospherically corrected Level-2A of S2 data can be accessed at the GEE repository. The acquisition time of S2 is between 10:11 and 10:22 AM local time for the study area. Nearly cloud-free products



captured over the study area were used: L8/L9 Operational Land Imager (OLI)/Thermal Infrared Sensor (TIRS) and Multispectral Instrument (MSI) onboard S2 satellites. Table 1 summarises the data acquisition dates, cloud cover, and local minimum and maximum air temperature from the Bureau of Meteorology (BOM).

**Table 1.** Acquisition dates for L8/L9 with the percentage of cloud cover of the scene, local minimum and maximum air temperature (Air Temp min/max), and the closest S2 images used for downscaling.

| Landsat 8/9           | Cloud [%] | Air Temp Min/Max [°C] | Sentinel-2       | Cloud [%] |
|-----------------------|-----------|-----------------------|------------------|-----------|
| 12 December 2023 (L8) | 0.04      | 17.2/34.6             | 12 December 2023 | 0.00      |
| 13 December 2023 (L9) | 0.01      | 17.3/30.4             | 12 December 2023 | 0.00      |
| 20 December 2023 (L9) | 5.00      | 19.2/37.1             | 22 December 2023 | 6.21      |
| 05 January 2024 (L9)  | 0.12      | 16.2/30.9             | 06 January 2024  | 0.00      |
| 06 January 2024 (L8)  | 0.01      | 17.7/33.4             | 06 January 2024  | 0.00      |
| 13 January 2024 (L8)  | 4.01      | 23.2/40.7             | 11 January 2024  | 6.00      |
| 21 January 2024 (L9)  | 0.12      | 19.0/30.3             | 21 January 2024  | 4.94      |

In addition to L8/9 data and S2 data, the Ecosystem Spaceborne Thermal Radiometer Experiment on Space Station (ECOSTRESS) data were used as a reference to validate the downscaling results. This mission shares similarities with L8/9 regarding spatial resolution and the overpass time being close to Landsat for some days. Mounted on the International Space Station (ISS), ECOSTRESS was launched by NASA in 2018 and measures thermal infrared radiance to provide detailed insights into the surface energy balance, with LST and emissivity [31]. The main mission is particularly to understand plant–water dynamics through evapotranspiration estimation, urban heat island effects, and drought monitoring [32]. ECOSTRESS offers a spatial resolution of approximately 70 m [33].

## 2.2.2. Ground-Based Measurements

The ground-based data collection aimed to evaluate the accuracy of the LST downscaling results. For this phase, we conducted measurements of surface temperature within the study area using Q1283A infrared non-contact thermometer devices. Five thermometers featuring laser-target pointers were used to collect temperature data. These instruments can record temperatures ranging from  $-50$  to  $580$  °C with a resolution of  $0.1$  °C and a basic accuracy of  $\pm 2\%$  of the reading. The spectral response of the sensor encompasses wavelengths between  $8$  and  $14$   $\mu\text{m}$ .

Systematic temperature recordings were performed at locations and times close to the satellite's overpass, spanning within an interval of  $-30$  to  $+30$  min of the satellite's passage. At each site, a single measurement was made. The sample locations were strategically chosen to represent various environmental settings such as roofs, grasslands, unirrigated grass, and asphalt covers. Due to practical measurement challenges, the classes representing water and trees were excluded. The ground samples were well distributed across the study area, as presented in Figure 1 with yellow dots. The exact locations of the sample sites were determined using Global Navigation Satellite System (GNSS) receivers. We successfully collected 357 in situ surface temperatures. A summary of the collected temperature data is presented in Table 2.

**Table 2.** Summary of the collected surface temperature in the study area: minimum surface temperature (Min Ts), maximum surface temperature (Max Ts), and average temperature (Av Ts).

| Land Cover Class  | Sample Count | Min Ts [°C] | Max Ts [°C] | Av Ts [°C] |
|-------------------|--------------|-------------|-------------|------------|
| Asphalt           | 98           | 35.60       | 50.70       | 43.59      |
| Grass             | 74           | 28.30       | 39.10       | 34.55      |
| Colorbond roof    | 57           | 35.00       | 54.80       | 44.60      |
| Tile roof         | 97           | 36.00       | 51.30       | 43.13      |
| Unirrigated grass | 31           | 36.10       | 53.20       | 43.02      |

Table 2 highlights the temperature variations across different land cover classes, providing valuable insights into the thermal characteristics of various surfaces within the study area. The differences in temperatures between and within each class can be attributed to the inherent properties of each surface type, such as heat absorption, emissivity, colour, heat reflection, and moisture retention. Colorbond roof (mostly with dark colour in the study area) was the hottest, followed by asphalt, tile roof, and unirrigated grass, while grassland recorded the coolest surface. Relatively small samples of unirrigated grass were collected due to the sparse occurrence of this land cover class in the study area. However, 31 samples are sufficient for the validation, as suggested by [34].

### 2.3. Methods

#### 2.3.1. Satellite Imagery Band Naming

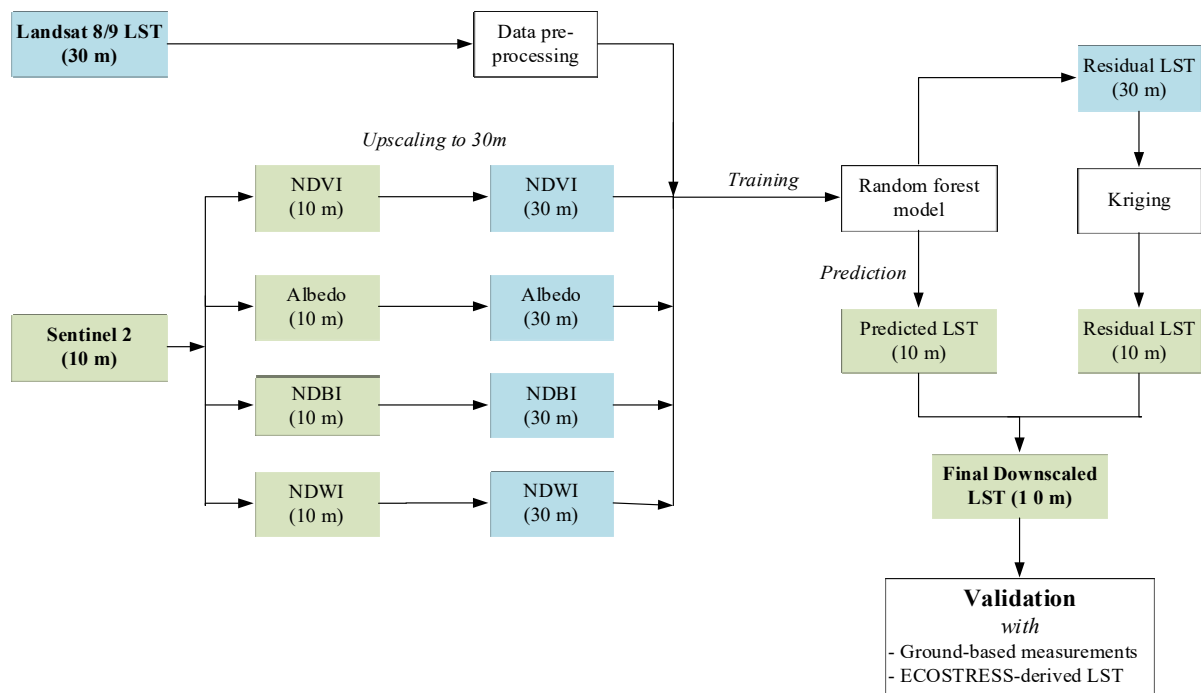
In the subsequent sections, the variables B, G, R, NIR, SWIR1, and SWIR2 denote the reflectance values corresponding to the blue, green, red, near-infrared, shortwave infrared 1, and shortwave infrared 2 bands. The script *v* indicates the reflectance value of each band.

#### 2.3.2. LST from Landsat 8/9 and Data Processing

The original LST is sourced from the National Aeronautics and Space Administration (NASA) website. The LST is derived from the thermal infrared (TIR) band 10 of Landsat 8 and 9, available as collection 2, already processed with atmospheric and emissivity correction, and resampled to 30 m by NASA. The flowchart of the proposed downscaling method (GeoML) is presented in Figure 2. The 30 m LST was used to train the random forest (RF) model after removing outliers or some pixels contaminated by clouds. In addition, we computed the coefficient of variance (CV) to improve the training data. It measures the homogeneity inside each coarse-resolution LST pixel [14]. The closer the CV is to 0, the purer the pixel is. The CV values were used as weights in our training, with the most homogeneous pixels weighted more and the least homogeneous discarded. A threshold was set depending on the data, usually leaving around 80% of the pixels available for training. This method was previously used by [15] with linear regression methods to ensure the selection of relatively uniform pixels—those with the lowest CV—rather than using all the pixels of the LST image. It was suggested that using pixels with a low CV improves the relationship between the LST and high-resolution independent variables [14,15]. The calculation of the coefficient CV is given in Equation (1).

$$CV() = \frac{SD()}{\text{mean}()} \quad (1)$$

where SD is the standard deviation, and the variable in brackets is the content of a moving window of  $5 \times 5$  pixels of an index image at 30 m.



**Figure 2.** Flowchart of the GeoML downscaling process of LST. The blue background indicates the low-resolution data, and the green one indicates the high-resolution data.

### 2.3.3. Temperature Predictors from Sentinel-2

S2 data were used to derive the high-resolution auxiliary data as independent variables in the LST downscaling process. The first variable is the surface albedo, a measure of the proportion of incoming solar radiation a surface reflects. The surface albedo is a dimensionless quantity with values ranging from 0 (no reflection) to 1 (total reflection). The values depend on surface properties such as material composition, moisture content, texture, and the angle of incidence of solar radiation [35]. For this study, the calculation of the surface albedo ( $\alpha$ ) is shown in Equation (2) [36].

$$\alpha = 0.2266v_B + 0.1236v_G + 0.1573v_R + 0.3417v_{NIR} + 0.1170v_{SWIR1} + 0.0338v_{SWIR2} \quad (2)$$

The three other independent variables are spectral indices, including the normalised difference vegetation index (NDVI), the normalised difference built-up index (NDBI), and the normalised difference water index (NDWI). The NDVI represents greenness and quantifies the health and density of vegetation [37]. It is computed as in Equation (3) and has values ranging from  $-1$  to  $+1$ ; negative values indicate water bodies or highly reflective surfaces; values close to 0 correspond to bare soil, urban areas, or sparse/stressed vegetation; and high values indicate dense and healthy vegetation.

$$NDVI = \frac{v_{NIR} - v_R}{v_{NIR} + v_R} \quad (3)$$

The NDBI is a spectral index that identifies and maps built-up and urban areas [38]. It is calculated as in Equation (4) and has values from  $-1$  to  $+1$ . Low negative values correspond to vegetation, water bodies, or bare soil, while high positive values represent built-up or impervious surfaces, such as asphalt and concrete [15].

$$NDBI = \frac{v_{SWIR1} - v_{NIR}}{v_{SWIR1} + v_{NIR}} \quad (4)$$

The NDWI is used to identify water bodies [39]. It is calculated after [40] in Equation (5) and has values between  $-1$  and  $+1$ . Negative values correspond to bare soil, vegetation, or built-up areas; near-zero values indicate areas with mixed water content, such as wetlands or moist soil; and positive values represent water, with purer water surface related to high values [39,40].

$$\text{NDWI} = \frac{v_G - v_{\text{NIR}}}{v_G + v_{\text{NIR}}} \quad (5)$$

With the four indices mentioned above, it is important to handle multicollinearity among the predictor variables of the model. The correlation between each pair of variables needs to be calculated, and correlated variables should be removed from the model. The package “ENMTools” within the R environment was used to compute the Pearson’s correlation coefficient among variables [41]. The absolute correlation values range from 0 (no relationship) to 1 (fully correlated) and can be negative or positive. The significance of the correlation was determined based on [42], and the statistically significant variables were removed.

#### 2.3.4. Random Forest Regression

Random forest (RF) is an aggregated predictor that uses hierarchical constraints to predict the features represented by the datasets. It uses a random bagging method to select all training subsets, and a decision tree is formed from every subset. The bagging method randomly resamples the original dataset into multiple training subsets. One-third of the total instances are randomly picked to validate the prediction accuracy rather than being used to create the model. As such, RF can provide an unbiased estimation of the error without using other data subsets. During the partition process, the algorithm proceeds through the optimal split, measured by the maximum reduction in impurity. There are many approximations for impurity measurements, with the most commonly measured by the Gini index, as shown in Equation (6) [43].

$$I_G t_{Y(X_i)} = 1 - \sum_{j=1}^m f(t_{Y(X_i)}, j)^2 \quad (6)$$

where  $f(t_{Y(X_i)}, j)$  is the share of samples for which the value of the feature  $X_i$  belongs to the leaf  $j$  and the node  $t(x)$ ,  $Y$  is the search value of the splitting process, and  $m$  represents the number of trees. The decision tree that splits the criteria selection is based on the lowest  $I_G$ . The result of RF modelling is determined by the average of all decision tree predictions [21] in Equation (7).

$$f_{rf}^K(X) = \frac{1}{K} \sum_{k=1}^K T(X). \quad (7)$$

where  $X$  is the input vector,  $K$  represents the number of regression trees, and  $T(X)$  denotes the result of the decision tree prediction.

The “ranger” package in the R environment was used to run the RF modelling. It is the fastest and most memory-efficient implementation of RF, using a C++ source package, for high-dimensional data [44].

Several parameters were considered in RF modelling. Firstly, one important parameter is the number of trees—a hyperparameter in RF that defines how many decision trees will be built. A larger number of trees often leads to better model performance, but at the cost of increased computational time. In practice, 500–1000 trees are commonly used, but the optimal number depends on the dataset and resources. For this study, 500 trees were used for the LST downscaling as this provided the highest performance and minimised the risk of the model overfitting. Secondly, there is the “mtry”, the number of variables



randomly chosen as candidates for splitting at each node in a decision tree. This random selection introduces variability among the trees, improving generalisation and reducing overfitting. The value of “mtry” is commonly calculated as the square root of the number of variables. For this study, it was set to 2. Thirdly, “min.node.size” is a parameter that controls the smallest number of observations (samples) allowed in a terminal (leaf) node of each decision tree. It was set to 5 for this modelling. Finally, the “Splitrule” parameter specifies the criterion for evaluating and choosing the best feature and threshold for splitting nodes in each decision tree. It determines how the trees will make splits during training, guiding the algorithm in choosing the best feature and cut-off for each node in a tree. “Variance” was used for our modelling to minimise the variance in the regression prediction.

In addition, the variable importance can also be set; it refers to the technique used to determine the contribution of each predictor variable to the model prediction. In other words, it tells which features influence the model’s output most and which are less relevant [45]. For our model, “impurity” was chosen, meaning that the importance of a variable is calculated by how much it decreases the impurity across all decision trees in the forest.

### 2.3.5. Error Estimation Using Kriging

The residual LST at coarse resolution (30 m spatial resolution for this study) is defined as the difference between the LST estimated using the RF model at 30 m and the observed L8/9 LST (30 m). This residual was downscaled to a high-resolution 10 m using ordinary kriging. Ordinary kriging can be considered an interpolation based on a weighted sum of known LST residual (30 m) values, where weights are derived from the semi-variogram model. The predicted high-resolution LST residual at 10 m is given by Equation (8).

$$Z^*(s_0) = \sum_{i=1}^N \lambda_i Z(s_i) \quad (8)$$

where  $Z^*(s_0)$  is the predicted residual LST value at the target location  $s_0$ ,  $Z(s_i)$  represents the known residual LST at 30 m resolution, and  $\lambda_i$  represents the kriging weights. They are determined by solving the kriging system, ensuring an unbiased estimate with minimum variance. Specifically, the weights are obtained by fitting a variogram model to the observed data. A variogram describes the correlation between observation values and the distance between locations. For this study, the Gaussian variogram model defined in Equation (9) was used.

$$\gamma(h) = n + p \cdot \left(1 - \exp\left(-\frac{h^2}{a^2}\right)\right) \quad (9)$$

where  $h$  is the lag distance between two points,  $n$  (nugget) is the variance at zero distance,  $p$  (psill) is the difference between the total sill and nugget, representing spatial correlation, and  $a$  is the range, a distance at which the variogram levels off, indicating the limit of spatial correlation. The final fine-scale (10 m) LST predictions were estimated by adding downsampled regression residuals using kriging to the predicted LST trend obtained from the RF model.

### 2.3.6. GeoML Downscaling

The 30 m LST derived from L8/9 was downsampled using the proposed method “GeoML”. Using the bilinear resampling method, the 10 m independent variables, including albedo, NDVI, NDBI, and NDWI, were upsampled to 30 m. Then, the random forest (RF) model was trained using the ranger package in R programming. The high-resolution LST was obtained by using the model prediction at 10 m. The 30 m residual LST was downsampled to 10 m using ordinary kriging and then combined with the high-resolution prediction from RF to obtain the results.

### 2.3.7. HUTS Downscaling

In this study, the high-resolution urban thermal sharpener (HUTS) downscaling method was also applied to downscale the same data, and the results were used as a reference for comparison. The HUTS model expressed the value of LST as a non-linear function: a 4th-order bivariate regression based on NDVI and the surface albedo. The downscaling consists of applying Equation (10) to the NDVI and albedo of each high-resolution pixel to obtain the predicted high-resolution LST.

$$T_{\text{LST}} = \sum_{k,m \geq 0, k+m \leq 4} p_{k,m} v_{\text{NDVI}}^k \alpha^m + \Delta T_{\text{LST}} \quad (10)$$

where  $T_{\text{LST}}$  is the land surface temperature,  $p_{k,m}$  represents the regression coefficients,  $\alpha$  is the value of the surface albedo, and  $\Delta T_{\text{LST}}$  is the residual of the regression. More details about the HUTS downscaling method can be found in the paper by Dominguez et al. [18].

### 2.3.8. Validation Method

To assess the performance of the proposed downscaling technique, we used a field-based approach with ground-based measurements and an image-based approach using LST-derived data from another sensor as references. For validation, we employed assessment metrics to quantify and analyse the disparities between the downscaled LST and the reference data. Pearson's correlation coefficient  $r$  provides the degree of correlation between the two datasets [12]. In addition, the root mean square error (RMSE) was employed to quantify the magnitude of the differences between the predicted (downscaled LST) and observed temperature (ground-based temperature) values [46]. RMSE provides a measure of the overall model accuracy. Lastly, the mean absolute error (MAE) was utilised to assess the average magnitude of the errors of the two mentioned variables [17]. These assessment metrics are calculated in Equations (11)–(13).

$$r = \frac{\sum_{i=1}^N (X_i - \bar{X})(Y_i - \bar{Y})}{\sqrt{\sum_{i=1}^N (X_i - \bar{X})^2 \sum_{i=1}^N (Y_i - \bar{Y})^2}} \quad (11)$$

$$RMSE = \sqrt{\frac{\sum_{i=1}^N (X_i - Y_i)^2}{N}} \quad (12)$$

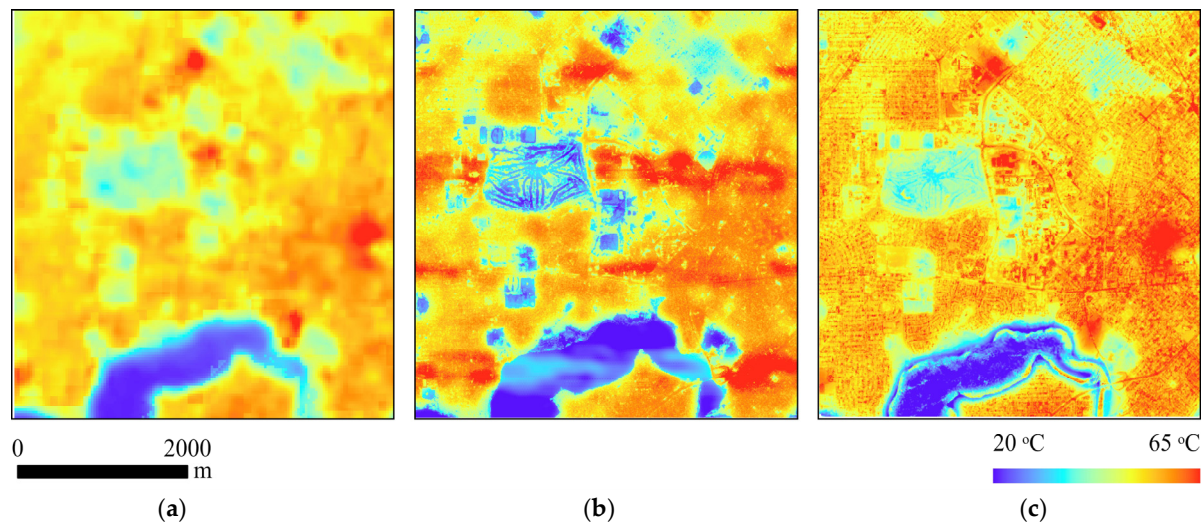
$$MAE = \frac{1}{N} \sum_{i=1}^N |X_i - Y_i| \quad (13)$$

where  $N$  is the total number of observations,  $X_i$  is the  $i$ -th in situ measurement data,  $\bar{X}$  is the mean value of  $X_i$ ,  $Y_i$  is the corresponding predicted LST data, and  $\bar{Y}$  is the mean value of  $Y_i$ .

## 3. Results and Validation

### 3.1. Results

The results of the downscaling of 30 m LST from L8 data observed on 12 December 2023, using the proposed GeoML approach and the HUTS method, are provided in Figure 3.



**Figure 3.** Downscaling results: (a) original LST 30 m, (b) enhanced LST 10 m using RF and kriging, and (c) enhanced LST 10 m using HUTS. The temperature–colour scale is the same for the three figures. Image captured on 12 December 2023.

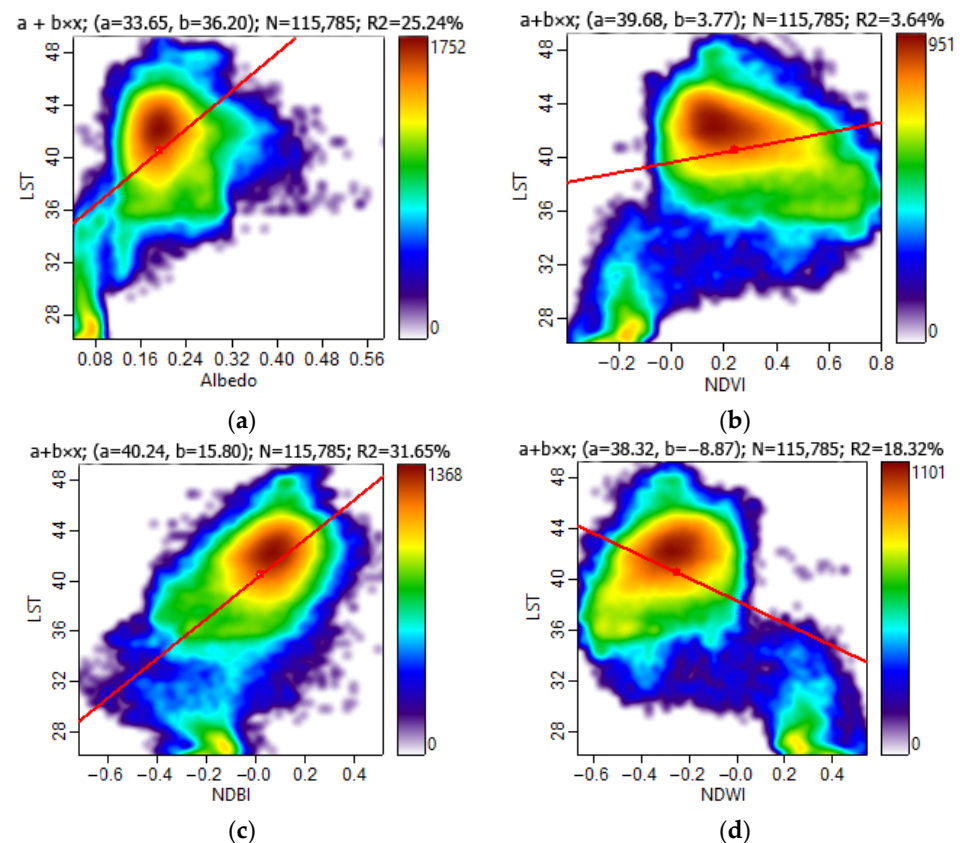
The results demonstrate that details related to the independent variables, indicated by the surface albedo and the spectral indices representing the land cover features, appear in the downscaled LST data. On the one hand, high LST values correspond to dense urban areas, mainly with impervious surfaces, such as buildings, roads, and areas covered with concrete. On the other hand, pixels with low LST values are related to areas covered by vegetation, such as green parks and forests, and wetlands, including water bodies and irrigated grassland. Major urban features, such as buildings, roads, and parking lots, are clearly depicted in Figure 3b, compared to the observed low-resolution LST in Figure 3a. However, small urban features, specifically those less than 10 m, are not well resolved due to the 10 m resolution of S2. For visual comparison purposes, the downscaled 10 m LST derived from HUTS methods is also presented in Figure 3c. While the HUTS model effectively captures urban thermal patterns and aligns well with urban characteristics, it tends to overestimate temperatures in areas covered by vegetation and irrigated features.

### 3.1.1. Multi-Collinearity Among Predictors

For the downscaling of the LST derived from satellite data on 12 December 2023, the summary of the correlation matrix among the four predictor variables derived from S2 is presented in Table 3. The absolute Pearson’s correlation coefficient values range from 0 (no relationship) to 1 (fully correlated) and can be negative or positive. Based on six pairs (of the four independent variables), all correlations less than or equal to  $-0.92$  or greater than or equal to  $0.92$  were deemed significant at  $\alpha = 0.01$  (significance level at  $p$ -value 0.01) and removed. Although there is a high-value negative correlation between NDVI and NDWI, it is not significant at  $\alpha = 0.01$ . Therefore, there is no significant collinearity issue among the independent variables, and all of them were used in the model for the LST data on 12 December 2023. Some independent variables were removed depending on the correlation coefficient for the data on different dates. The correlation between the LST and the four independent variables is represented in Figure 4 for the LST data captured on 12 December 2023. A high positive correlation is observed between the pairs LST–albedo (Figure 4a) and LST–NDBI (Figure 4c), a weak positive correlation between LST and NDVI (Figure 4b), and a negative correlation between LST and NDWI (Figure 4d).

**Table 3.** Correlation matrix among the four independent variables.

|        | NDVI   | Albedo | NDBI   | NDWI   |
|--------|--------|--------|--------|--------|
| NDVI   | 1.000  | 0.222  | −0.348 | −0.912 |
| Albedo | 0.222  | 1.000  | 0.114  | −0.383 |
| NDBI   | −0.348 | 0.114  | 1.000  | 0.109  |
| NDWI   | −0.912 | −0.383 | 0.109  | 1.000  |

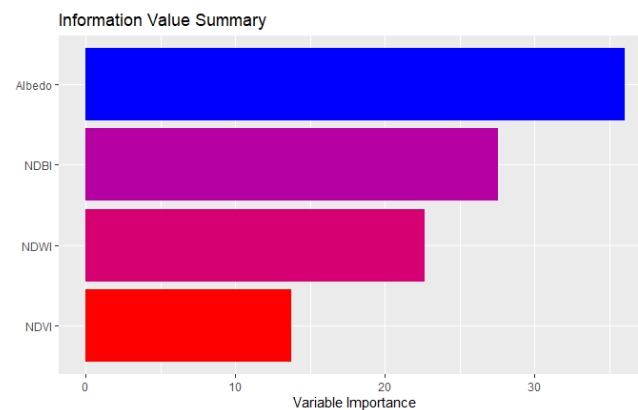
**Figure 4.** Scatterplots showing the correlation between spectral indices and LST: (a) LST and albedo, (b) LST and NDVI, (c) LST and NDBI, (d) LST and NDWI. Data from imagery captured on 12 December 2023. The value in the legend indicates the number of points per pixel.

### 3.1.2. Random Forest Modelling

The RF downscaling experiments were implemented in the R program using the ranger package [44]. Several parameters, including the number of trees, the number of randomly selected variables at each split, the minimum number of observations per terminal node, and the criteria determining how nodes were split, were experimented with during the modelling.

The result of variable importance for the data on 12 December 2023 is presented in Figure 5, where the rank of the variable importance is highlighted. In terms of percentage, the contributions of surface albedo, NDBI, NDWI, and NDVI are 27.58%, 22.66%, 13.74%, and 6.02%, respectively. In Figure 5, the contribution of independent variables gradually decreases. The surface albedo has the highest importance among all the variables, indicating a high correlation between the LST and albedo (Figure 4a), which is explained by the massive presence of dark and reflective roofs and asphalt in the study area. NDBI was the second most important predictor, which is also highly correlated to LST due to the high proportion of built-up area. The least important variables are NDWI and NDVI due to

the low correlation between these indices and the LST, which can be explained by the low abundance of water and sparse vegetation in the study area.

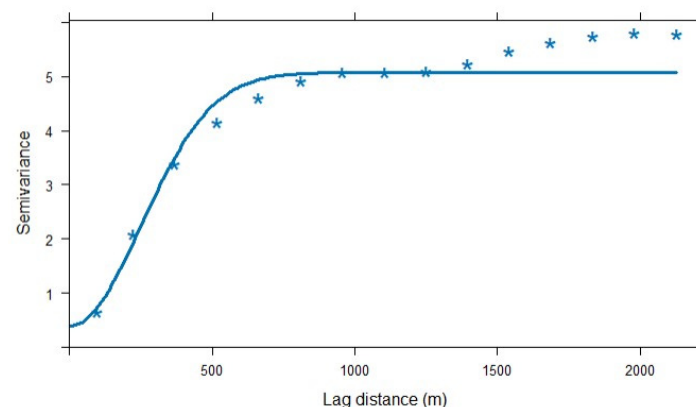


**Figure 5.** Random forest ranking importance of the independent variables.

For the model performance, the out-of-bag (OOB) prediction mean square error (MSE) was 2.013; it measures the model error based on the data points left out during the training of each tree. It is a form of cross-validation that estimates the model's generalisation ability without requiring a separate test set. The R-squared (OOB), which estimates how well the RF model predicts the out-of-bag (OOB) samples, was 0.631. It is a cross-validation-like metric that indicates how much variance in the target variable is explained by the model using only the training data.

### 3.1.3. Kriging of the Residual

Several experiments were conducted to find the best-fitted model in the 'gstat' package, with a nugget variance of 0.1, partial sill of 1.6, and range of 1000. We then used the Gaussian variogram to predict the residuals at fine resolution using the ordinary kriging method in a neighbourhood with 50 nearest observations. The variogram model (blue point) and the fitted variogram (blue line) for the data on 12 December 2023 are shown in Figure 6.



**Figure 6.** Variogram model and fitted variogram. The symbols (\*) represent the empirical variogram model values, while the solid line denotes the fitted variogram.

### 3.2. Validation of the Downscaled LST Data

Using the surface temperature data collected from ground-based measurements as described in Section 2.2.2, the LST validation (field-based validation) resulting from the GeoML downscaling process was carried out across the entire study area and by land cover classes. Moreover, a cross-platform validation (image-based validation) was also performed for the entire study area using observed LST from ECOSTRESS as a reference image.



### 3.2.1. Field-Based Scene Validation

The result of validation within the study area is presented in the first row of Table 4. This table presents Pearson's correlation coefficient  $r$ ,  $RMSE$ , and  $MAE$  between the field observations as reference and LST. In Table 4, there are three types of LST: LST-30 m is the observed low-resolution LST from L8/9, HUTS-10 m refers to the downscaled LST using the HUTS model, and GeoML-10 m corresponds to the downscaled LST using the proposed GeoML. The results demonstrate a high correlation between the downscaled LST and the ground-based measurements. Based on the validation across the entire study area, GeoML performs better than HUTS in terms of  $r$  (0.850 vs. 0.834),  $RMSE$  (2.708 °C vs. 2.966 °C), and  $MAE$  (2.197 °C vs. 2.487 °C). These findings suggest that combining random forest and kriging can better handle the complex relationship between LST and heterogeneous surface characteristics than HUTS.

**Table 4.** Difference between surface temperature from ground measurements and LST derived from satellite (LST-30 m) and enhanced data with HUTS (HUTS-10 m) and with GeoML (GeoML-10 m).

| Model             | LST-30 m |                |               | HUTS-10 m |                |               | GeoML-10 m   |                |               |
|-------------------|----------|----------------|---------------|-----------|----------------|---------------|--------------|----------------|---------------|
| Land Cover Class  | $r$      | $RMSE$<br>[°C] | $MAE$<br>[°C] | $r$       | $RMSE$<br>[°C] | $MAE$<br>[°C] | $r$          | $RMSE$<br>[°C] | $MAE$<br>[°C] |
| Entire image      | 0.814    | 3.030          | 2.490         | 0.834     | 2.966          | 2.487         | <b>0.850</b> | <b>2.708</b>   | <b>2.197</b>  |
| Asphalt           | 0.487    | 2.841          | 2.287         | 0.490     | 2.92           | 2.487         | <b>0.586</b> | <b>2.521</b>   | <b>2.037</b>  |
| Grass             | 0.485    | 3.445          | 2.873         | 0.582     | 3.230          | 2.769         | 0.572        | <b>2.305</b>   | <b>1.713</b>  |
| Colorbond roof    | 0.840    | 2.599          | 2.120         | 0.845     | 2.580          | 2.081         | 0.840        | 2.837          | 2.375         |
| Tile roof         | 0.494    | 3.076          | 2.581         | 0.536     | 2.994          | 2.478         | 0.517        | 3.016          | 2.507         |
| Unirrigated grass | 0.777    | 3.135          | 2.608         | 0.789     | 3.074          | 2.645         | <b>0.795</b> | <b>2.908</b>   | <b>2.565</b>  |

### 3.2.2. Field-Based Validation per Land Cover Class

The validation with the same ground-based data, but categorised by land cover class, is presented in Table 4. For this study, five classes, namely asphalt, grass, Colorbond roof, tile roof, and unirrigated grass, were chosen during the field data collection. Table 4 presents the validation results in row per land cover class. Similar to the entire image of the study area,  $r$ ,  $RMSE$ , and  $MAE$  were calculated within each land cover class. The results indicate that the correlation coefficient between the in situ measurements and downscaled LST by GeoML is notably higher than the one obtained with HUTS for asphalt land cover. However, the correlation values are comparable for the remaining land cover classes, with those of HUTS slightly higher. Regarding  $RMSE$  and  $MAE$ , GeoML provided higher accuracy for asphalt, grass, and unirrigated grass than HUTS; however, HUTS outperformed GeoML for Colorbond and tile roofs.

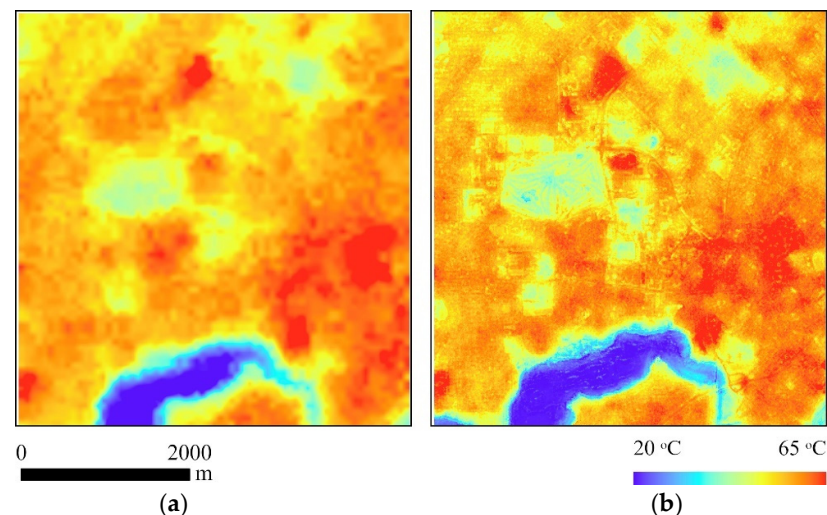
### 3.2.3. Image-Based Validation

Due to the absence of 10 m or higher spatial resolution LST, such as aerial or airborne thermal data, as a ground truth reference, a cross-platform validation was conducted to further validate downscaled LST data. ECOSTRESS LST-derived data were used as reference images for the entire study area. Although ECOSTRESS has a native spatial resolution of approximately 70 m, it was the only available satellite-based LST product with acquisition times closely aligned with the L9 overpass. Specifically, the ECOSTRESS image acquired on 5 January 2024 at 10:28:19 a.m. was selected for validation, as it was temporally close to the L9 overpass at 10:12:08 a.m. on the same day. Despite the spatial resolution disparity, this ECOSTRESS scene offered the best available proxy for assessing

the spatial pattern and magnitude of the downscaled LST. Table 5 indicates the validation results of the downscaled LST (Figure 7b) with GeoML and the observed LST derived from ECOSTRESS (Figure 7a). The table indicates the Pearson's correlation coefficient  $r$ , the  $RMSE$ , and the  $MAE$  between the ECOSTRESS LST and the observed Landsat LST before downscaling in the second column, and the ECOSTRESS LST and the downscaled LST with GeoML in the third column. Table 5 shows a high correlation between the LST derived from ECOSTRESS and both the observed L9 LST (30 m) and the enhanced L9 LST (10 m). There is a gradual improvement in correlation and errors after the downscaling process.

**Table 5.** Validation results showing the comparison between the observed (original) LST and ECOSTRESS LST (second column), and between the enhanced LST and ECOSTRESS LST (third column).

|             | Original LST | Enhanced LST |
|-------------|--------------|--------------|
| $r$         | 0.928        | <b>0.937</b> |
| $RMSE$ [°C] | 1.736        | <b>1.719</b> |
| $MAE$ [°C]  | 1.245        | <b>1.237</b> |



**Figure 7.** Validation with ECOSTRESS: (a) ECOSTRESS LST resampled to 10 m, (b) downscaled LST (10 m) from L9 data. Image captured on 05 January 2024.

#### 4. Discussion

This study aimed to improve the spatial resolution of LST derived from freely available L8/9 data using high-resolution Sentinel-2 datasets for urban studies. The proposed approach suggests an estimation of LST at a fine scale: the 30 m LST was downscaled to a fine spatial resolution (10 m) using auxiliary variables from visible and near-infrared (VNIR) data. Visual analysis shows that the downscaling results are consistent with the observed LST and reflect the land cover characteristics indicated by the high-resolution spectral indices and surface albedo derived from Sentinel-2. Additionally, the validation results demonstrate that the enhanced LST is in good agreement with both ground-based measurements and the observed LST derived from ECOSTRESS, based on the validation metrics. Therefore, the downscaling methods can be considered satisfactory, making the LST data suitable for environmental analyses related to the UHI effect and other applications requiring detailed LST data at fine scales.

Despite the strong agreement between the downscaled LST and the surface temperature obtained from ground-based measurements, some significant differences exist in certain areas. The highest difference, approximately 7 °C, between the values recorded

from in situ measurements and the enhanced LST generated by GeoML can be observed in a few sample areas, mainly roofs, grass, and asphalt. These differences may be a result of variations in measurement approaches used to record these values. While in situ measurements read the temperature at a point level, satellites record an aggregate value of the temperature of each land cover component of a given pixel, showing a mixed effect in heterogeneous areas. The LST of different surfaces, such as concrete and roofs, comprises the resulting LST with 10 m resolution, generating an average value. However, the differences are smaller in homogeneous areas with the same material within the grid. Thus, the largest difference occurs in cells with aggregated surface materials. However, they only occur in a few locations, as most ground measurements were collected in homogeneous areas.

While earlier studies (e.g., [47,48]) only used the normalised vegetation index (NDVI) as a temperature predictor to downscale LST, this study, in agreement with other recent studies (e.g., [49,50]), suggested that incorporating temperature predictors in addition to NDVI has the potential to improve the accuracy of spatial resolution enhancement in complex landscapes, especially in cities with sparse urban forests. The findings of this study show that the inclusion of surface albedo, NDBI, and NDWI improved the performance and the accuracy of the models, compared to the traditional use of NDVI alone in the downscaling process. Earlier studies indicate that using independent variables that strongly correlate with LST improves the model [51,52]. However, multicollinearity should be checked among highly correlated independent variables, as correlated variables affect the model's performance. For RF modelling, the contribution of each variable can be ranked, which is highly important in choosing the predictors to use in a model. This process is important for optimising the selection of independent variables and creating a final subset from all candidates to lower dimensionality, reduce the data redundancy and collinearity, and improve the efficiency of the downscaling process [53]. This practice was used in this study to improve the results.

The proposed GeoML method performs better than the HUTS model based on the assessment against the ground-based measurements in the entire study area. This superiority stems from GeoML's ability, as a non-linear regression RF model, to capture the complex relationship between LST and temperature predictors while incorporating spatial dependency through kriging-based residual downscaling. In addition, including two extra independent variables—NDBI and NDWI—enhances GeoML's performance by accounting for built-up areas and the presence of water bodies. The variable importance analysis reveals that NDBI contributes more significantly than NDVI, likely due to the low vegetation cover in the study area. This finding aligns with previous studies (e.g., [16,49]), which suggests that NDVI is a less effective temperature predictor in urban environments where built-up areas dominate. These results also reinforce established urban thermal dynamics principles, highlighting that various land cover types influence heating and cooling differently. Areas covered by impervious surfaces, such as buildings and roads, are the main contributors to high LST. In contrast, vegetation helps reduce it through its cooling effect. However, certain impervious areas may exhibit lower LST due to the cooling influence of shadows.

Although the GeoML approach demonstrates superior accuracy compared to HUTS when validated against ground-based measurements, the magnitude of improvement remains relatively modest. This marginal gain must be weighed against the higher computational cost of the GeoML framework. Based on experimental implementation, the average runtime for downscaling one Landsat data covering the study area using GeoML, including model training, variable selection, and kriging-based residual correction, was approximately 25 min on a standard desktop with 32 GB RAM and a 12-core CPU. In contrast, the HUTS method completed the same task in about 10 min under the same computing

environment. The simplicity of HUTS makes it more computationally efficient and easier to deploy for large-scale or operational monitoring. While GeoML provides enhanced spatial accuracy, the improvement over HUTS should be critically evaluated in the context of application-specific needs, especially where processing resources or near-real-time output are key constraints.

The findings also indicate that both methods perform better in densely vegetated areas than in regions with highly heterogeneous impervious surfaces, suggesting that kriging is more effective in complex urban environments. Additionally, visual analysis (Figure 3) indicates that GeoML enhances the spatial details of LST while preserving intrinsic information and spatial patterns. However, the absence of an inner model means that localised extreme heat sources, such as factories emitting significant heat, may not be distinctly captured within the broader spatial framework.

## 5. Conclusions

This study confirms that enhancing the spatial resolution of LST using a hybrid geospatial machine learning (GeoML) framework, comprising random forest regression and kriging-based residual downscaling, is a viable and effective approach for urban thermal analysis. The integration of multiple predictor variables, such as NDVI, NDBI, NDWI, and surface albedo, yielded improved accuracy over traditional models, particularly in urban environments characterised by low vegetation cover.

The proposed GeoML method consistently outperformed the HUTS model in accuracy metrics, reinforcing the value of using non-linear models and geostatistical tools to capture complex urban thermal dynamics. However, the high computational demand of GeoML presents practical limitations, especially in large-scale or operational applications where rapid processing is essential. Consequently, the trade-off between accuracy and efficiency should be carefully considered in application-specific contexts.

Despite the good performance and high accuracy of the proposed downscaling approach, there is room for improvement, particularly in delineating smaller urban features. Future work should investigate the use of auxiliary data with higher spatial resolution (<10 m) from other optical sensors (e.g., Worldview-3 data) to resolve the temperature variations of smaller urban features such as buildings and narrow roads. Moreover, incorporating more sophisticated algorithms, such as deep learning with spatial features (e.g., geographically weighted deep neural networks), with other factors related to building, such as building density, radiation flux, digital elevation model (DEM), and other parameters related to surface heating and cooling such as thermal conductivity and evapotranspiration, may further enhance spatial accuracy.

Lastly, to strengthen the reliability of validation, future studies should consider deploying a dense network of continuously operating temperature sensors within the study area to provide more robust reference data for evaluating the downscaled LST.

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